Varesh Patel

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Education

Rutgers University **September 2025 - May 2027(Expected)**
Masters in Computer Science *New Brunswick, NJ*

G.H. Patel College of Engineering and Technology **September 2021 - April 2025**
B.Tech in Computer Science and Design *Anand, Gujarat, India*

Work Experience

Foundation for Ecological Security **January 2025 - May 2025**
Artificial Intelligence Intern

- Developed an Al bird classification system for IBIS 2.0, managing a 710GB image dataset (1,150 species).
- Achieved 79% accuracy with a two-stage deep learning pipeline. Deployed with GUI/CLI via Gradio for researchers.

Arcascope **October 2024 - January 2025**
Artificial Intelligence Intern

- Developed a 20-version machine learning pipeline for fatigue prediction, achieving 85% accuracy and 0.25 MAE through novel feature engineering and Runge-Kutta integration.
- Reduced MAE by 78% (from 1.17 to 0.25) with a custom 5+ layer PyTorch model and enhanced prediction accuracy from wearable data using a CORAL layer and specialized ordinal loss functions.

Projects

Emotion-Driven Spotify Clone

- Built a facial emotion recognition system using CNNs that achieved 95.83% accuracy.
- The system enables personalized music recommendations based on emotions, which boosted simulated user engagement by 25%.

Al-Powered Bird Species Classification

- Developed a classification system to enhance India's national biodiversity database by engineering a data pipeline to process a 710GB image dataset of 1,150 species.
- The model achieved 79% validation accuracy.

Daily Fatigue Prediction from Wearable Data

- Architected a machine learning pipeline that achieved 85% accuracy and a 0.25 Mean Absolute Error (MAE).
- Reduced MAE by 78% by implementing a deep ordinal regression model and engineering 9+ features.

PDFhelper - Document Management and Intelligent Retrieval App

- Developed a document management app allowing users to interact with PDFs.
- Implemented semantic document retrieval using Hugging Face embeddings and a Qdrant vector database, integrating LLaMA 3.2 to improve retrieval efficiency by 70%.

Skills

Programming Languages: Python, Java, C, C++
Frameworks & Libraries: PyTorch, TensorFlow, Scikit-learn, Keras, Pandas, NumPy, OpenCV
Tools & Technologies: Git, GitHub, Docker, Hugging Face

Leadership and Activities

Secretary, IEEE GCET Student Branch **December 2023 - December 2024**

- Organized 39+ technical and networking events, managed all administrative duties, and served as the primary communication link for the branch.

Publications

- Patel, V., Shah, K., & Joshi, K. (2025). ***Openpose, posenet and movenet: The evolution of deep learning methods in yoga pose and classification***. *6th International Conference on Data Analytics and Management (ICDAM-2025), London, United Kingdom.* (Publication in progress)
- Patel, V. (2024).***Machine learning-based hand gesture recognition for immersive gaming.*** *International Journal for Innovative Research in Multidisciplinary Field*, 10(56), 16-22.
- Patel, V., & Mehta, K. (2024). ***Emotion-aware music recommendations: Evaluating custom CNN vs. VGG16 and inception V3***. *2024 IEEE International Conference on Future Machine Learning and Data Science (FMLDS), 499-504.*